

REPORT 3 - Digital Forensics

This report aims to analyze the mobile phone of Heisenberg, detained on 2021-07-20, allegedly accused of dealing in stolen cars.

The first step is to produce a hash of the .zip file containing the phone's file system. Then, compare it with the hash given by the police to verify that no modification has been made throughout the custody chain.

The hash expected to get is: D67D931700D3C8A8343FD3A2B081AFADD3C2A75D

The hash I got: D67D931700D3C8A8343FD3a2B081AFADD3C2A75D

The hashes match, I can continue working.

The tool used to parse the data is: ALEAPP v3.5.0 which can be downloaded here:

<https://github.com/abrignoni/ALEAPP/releases/tag/v3.5.0>

And the specs of the system used to analyze the evidence is:

```
PRETTY_NAME="Ubuntu 24.04.4 LTS"
NAME="Ubuntu"
VERSION_ID="24.04"
VERSION="24.04.4 LTS (Noble Numbat)"
VERSION_CODENAME=noble
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
"
UBUNTU_CODENAME=noble
LOGO=ubuntu-logo
```

I parsed the .zip file using ALEAPP and generated an HTML report, which is included with this report. The hash value of the .zip file containing the ALEAPP report is:

3787360953608B6FFBA20158E4ADF1797D1969AF

In the next section, I answer the questions posed by the investigators and the court using the tool report and manual analysis:

Are there clues or evidence for Heisenberg dealing with (stolen) cars?

Non-definitive clues (meaning they are not clear clues that indicate that he was dealing with stolen cars, but induce some type of suspicion):

- **Reddit account:** With the name HeisenbergCarro, where “carro” is a word that can be used to talk about a car in Spanish, especially in Latin countries.
- **Email account:** There is also an email account called “heisenbergcarro@gmail.com” where the word “carro” appears again.
- **Bluetooth connections:** In the Bluetooth section, we identify three devices:

1st one (34:C7:31:F8:61:3B) (2021-04-05 03:15:28):

Called “CAR MULTIMEDIA”, clearly related to a car sound system. Even more, a MAC address lookup shows that the device was made by “ALPS ALPINE CO., LTD”, a multinational Japanese tech firm well known for making electronic and cooling components for cars. I have not been able to find more information regarding the possible model of the car. **However**, there are some things to keep in mind:

- Bluetooth names and MAC addresses can be spoofed or changed by the owner. It does not require extensive technological knowledge to do so. It is left to the jury to decide if this option is feasible.
- Even if it was not changed, this also does not prove that the car in question was stolen. This could be checked by contacting the manufacturer, providing the MAC address, and following the owners to see where that leads. That task is left to the investigators and for the jury to decide.

2nd one (2C:6B:7D:1D:21:A7) (2021-05-21 01:18:48):

Called “Bose SoundTouch CE2D10”, this Bluetooth device does not clearly map to being from a car. Nevertheless, after some basic searches for the name on Google, it is possible to find a company called “Bose” which does produce speakers, audio systems, etc. Some of them are directly related to cars and are sold in well-known and well-established companies: Porsche, Mitsubishi, Cadillac, Ford, etc. The MAC address in this case is issued by “Texas Instruments”, a global semiconductor giant known for manufacturing analog and embedded processing chips used in nearly all electronic systems. I’ve been able to find some evidence online matching this relationship, for example, Texas Instruments providing components for Bose. **IMPORTANT** things to keep in mind:

- All this evidence suggests that the names and MAC address have likely not been altered, but it is important to keep in mind that **IT IS AN OPTION**.
- This still does not prove that the cars connected were stolen.

3rd one (68:A8:E1:B2:B4:F7) (2021-07-09 05:45:28):

Called “SPEN 01 (B4F7) ZK”, the name matches a Samsung Pen and the MAC address matches a company called “Wacom Co., Ltd.”, known for making digital pens. So, it is not related to cars.

- **Car marketplace:** He has a lot of emails from a car marketplace app. This also does not mean he stole cars. However, this platform could be used to either sell stolen

cars or locate new victims, allowing him to choose the model to steal and contact the victim to get more information (**all of this is speculative**).

- **SMS:** Some SMS messages have been found where Heisenberg maintains a conversation in which he tries to sell a car. He mentions “stock” as a way of indicating that he has various vehicles available. He also asks for the reference that brought the buyer to contact him, justifying that he is trying to be cautious. Again, this does not confirm that the cars were stolen; it only proves that he was dealing with cars, stolen or not. Further down in this report, there is a conversation attached that shows this type of behaviour.
- **Images:** Numerous images of cars have been found with different edits changing their wheels.



- **Apps:** There are also multiple apps installed that are directly or indirectly related to dealing with cars:
 1. **CARFAX:** An application to check used car details, known accidents, damage, etc.
 2. **Venmo:** An application to make payments, which could have been used to close deals.
 3. **Orto:** An application used to check license plate information.
 4. **Facebook:** A well-known social media platform with the addition of **Facebook Marketplace**, where people can buy and sell items “non-officially”.
 5. **Cartomizer:** An application that lets you view different types of wheels on your car, such as those seen in the previous bullet point.

To sum up, there IS **NOT** clear and definitive proof that he was dealing with **STOLEN** cars. However, there are numerous facts that lead one to think that he was dealing with cars, stolen or not.

Heisenberg used his cellphone during the arrest. Did he create any recordings of the arrest? Note: arrest took place on 2021-07-20

There were a lot of calls during the day of the arrest, but all of them had a duration of 0, which is unusual. However, it does not seem like he could have recorded the arrest that way.

There is a video with the timestamp 2021-07-20 19:04:10. I am not aware of the exact time of the arrest, but if it matches, it is probably a video of it. It only has a duration of 2 seconds, during which it is possible to hear a sound resembling that of a door opening. It is important to mention that the video was found in the trash. The video will be included with the report; the SHA-1 hash of the video is: 5f6fd185e70ae40859cd4dbd2135255ffd586594

There is another video which is clearly from the arrest. In the video, it is possible to hear Heisenberg having a conversation with a woman apparently interested in buying the car.

After some questions about the vehicle, the woman confronts Heisenberg, stating that the vehicle has been reported stolen, and arrests him. The video will be included with the report; the SHA-1 hash of the video is: 9cf0d4ae0ad078460d1fc69b8ef372aeb7c942b2

Are there any hints that the police did not quite follow best practices in handling the cellphone during/after the arrest? If so, what did they do wrong?

There are a lot of app launches, usages, and connections after the official arrest. Some of the apps are: Twitter, Instagram, AndroidApp, Settings, etc.

This type of behaviour can occur under two types of circumstances. First, there is someone actively using the phone, which would mean a complete violation of the custody chain and could make all the evidence found on the telephone inadmissible in court.

The other scenario is that those accesses and events are produced automatically by the mobile phone. When a phone is seized, it should be isolated from external communications by turning on airplane mode, disabling Wi-Fi, Bluetooth, and any type of connectivity, and, if possible, placing it inside a Faraday bag. None of these measures was taken.

Are there any clues for Heisenberg taking an interest in cryptocurrency?

He has emails regarding crypto, e.g., Dogecoin, and Twitter messages in his feed also regarding crypto. Given that, especially on Twitter, the feed you receive is based on your interests, it is fairly reasonable to believe that he does have an interest in cryptocurrency.

Did Heisenberg use any apps for file hiding or encryption?

There are signs of Heisenberg using Signal, a privacy-first messaging app. In the last section of this report, it is possible to see those uses.

Are there any clues for Heisenberg connecting or planning to connect external drives to his Cellphone?

Yes, there are searches that indicate that he was interested in USB and whether they could perform specific actions on his device.

2021-05-24 17:54:47	can a samsung micro usb connector transfer data	https://www.google.com/search?q=can+a+samsung+micro+usb+connector+transfer+data&ro+usb+connector+trans&aqs=chrome.1.69i57j33i22i29i30.12274j0j9&client=ms-android-sa
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Was the image with the MD5 hash “066858f4b1971b0501b9a06296936a34” hidden by Heisenberg? If yes, what app was used?

By executing the command:

```
“find . -type f -exec md5sum {} \; | grep 066858f4b1971b0501b9a06296936a34”
```

under the output folder of ALEAPP I have identified an image with the exact MD5 hash:



It was hidden using an application called HideX, an app to hide files disguised as a calculator. This is known due to the fact that the image was found under a folder called “com.flatfish.cal.privacy”, which is the package name for that application.

Where did Heisenberg plan to meet with the owner of the telephone-number “+15402993169”?

Thanks to recovered SMS messages, it can be confirmed that Heisenberg was planning to meet with the owner of the telephone number (+15402993169) at Washington Street Tennis Courts at 18:42:55 on 07-20-2021.

Screenshots of the conversation between Heisenberg and +15402993169:

	1	Sent	Hey, thanks for calling, I will look up the inventory and get back to you in a few mins.
2021-07-19 22:15:43	1	Received	Great!
	1	Sent	Just to make sure, you wanted a Hyundai right?
2021-07-19 22:16:37	1	Received	Yes that is right! The newer the better.
	1	Sent	And just to be safe, can you tell me where you got my reference from?
2021-07-19 22:18:36	1	Received	Oh, it was one of my friends down in North Carolina who saw your Craigslist postings and mentioned it to me.
	1	Sent	Uhh ok, I guess I can work with that.
2021-07-19 22:20:05	1	Received	👍
	1	Sent	So, I checked and I still have a Hyundai Sonata, its probably a 2015, would you be interested?
2021-07-19 22:36:02	1	Received	Sure! Send me pictures and some info
	1	Sent	120K miles, 2014 Hyundai Sonata.
	1	Sent	Can't give you the VIN due to privacy. You can have it when you come over to see the vehicle.
2021-07-20 01:02:05	1	Received	Ok that sounds good!
	1	Sent	When would you like to meet to look at the car and seal the deal?
2021-07-20 01:20:22	1	Received	I will send you a text tomorrow morning and let you know what time tomorrow works best if that is ok with you?
	1	Sent	Sure, let me know, I will let you know if I am available or not. See you!

2021-07-20 18:42:32	1	Received	Can you meet in about 20mins at the Washington Street Tennis Courts?
	1	Sent	Sure, will see you at the building left of the courts.
2021-07-20 18:44:56	1	Received	Sounds good! See you!
	1	Sent	Hey Beth!

At which time(s) did Heisenberg actively use the Signal app on 2021-07-14?

It is possible to confirm the usage of Signal, labeled as “thoughtcrime.securesms”, on the following timestamps thanks to the tool ALEAPP and the module “Turbo - Application Usage”. It has also been possible to retrieve a screenshot of an usage report using the same tool.

2021-07-14 18:40:10.977	org.thoughtcrime.securesms
2021-07-14 18:40:10.977	org.thoughtcrime.securesms
2021-07-14 18:40:10.977	org.thoughtcrime.securesms
2021-07-14 18:39:52.149	org.thoughtcrime.securesms
2021-07-14 18:39:52.149	org.thoughtcrime.securesms
2021-07-14 18:39:52.149	org.thoughtcrime.securesms
2021-07-14 18:39:50.774	org.thoughtcrime.securesms
2021-07-14 18:39:50.774	org.thoughtcrime.securesms
2021-07-14 18:39:50.774	org.thoughtcrime.securesms

